



SEMITRANS® 3

Trench IGBT Modules

SKM600GB126D

Features

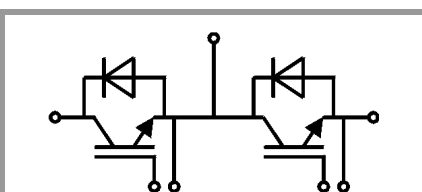
- Trench = Trenchgate technology
- $V_{CE(sat)}$ with positive temperature coefficient
- High short circuit capability, self limiting to $6 \times I_C$
- UL recognized, file no. E63532

Typical Applications*

- AC inverter drives
- UPS
- Electronic welders

Remarks

- $I_{DC} \leq 500A$ for $T_{Terminal} = 100^\circ C$



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Absolute Maximum Ratings				
Symbol	Conditions		Values	Unit
IGBT				
V _{CES}	T _j = 25 °C		1200	V
I _C	T _j = 150 °C	T _c = 25 °C	660	A
		T _c = 80 °C	461	A
I _{Cnom}			400	A
I _{CRM}	I _{CRM} = 2xI _{Cnom}		800	A
V _{GES}			-20 ... 20	V
t _{psc}	V _{CC} = 600 V V _{GE} ≤ 15 V V _{CES} ≤ 1200 V	T _j = 125 °C	10	μs
T _j			-40 ... 150	°C
Inverse diode				
I _F	T _j = 150 °C	T _c = 25 °C	490	A
		T _c = 80 °C	337	A
I _{Fnom}			400	A
I _{FRM}	I _{FRM} = 2xI _{Fnom}		800	A
I _{FSM}	t _p = 10 ms, sin 180°, T _j = 25 °C		3312	A
T _j			-40 ... 150	°C
Module				
I _{t(RMS)}			500	A
T _{stg}			-40 ... 125	°C
V _{isol}	AC sinus 50 Hz, t = 1 min		4000	V

Characteristics						
Symbol	Conditions		min.	typ.	max.	Unit
IGBT						
V _{CE(sat)}	I _C = 400 A	T _j = 25 °C		1.70	2.12	V
	V _{GE} = 15 V chipelevel	T _j = 125 °C		2.02	2.46	V
V _{CE0}	chipelevel	T _j = 25 °C		1	1.2	V
		T _j = 125 °C		0.9	1.1	V
r _{CE}	V _{GE} = 15 V chipelevel	T _j = 25 °C		1.75	2.3	mΩ
		T _j = 125 °C		2.8	3.4	mΩ
V _{GE(th)}	V _{GE} =V _{CE} , I _C = 16 mA		5	5.8	6.5	V
I _{CES}	V _{GE} = 0 V	T _j = 25 °C			5	mA
	V _{CE} = 1200 V	T _j = 125 °C				mA
C _{ies}	V _{CE} = 25 V V _{GE} = 0 V	f = 1 MHz		28.8		nF
C _{oes}		f = 1 MHz		1.51		nF
C _{res}		f = 1 MHz		1.31		nF
Q _G	V _{GE} = - 8 V...+ 20 V			3600		nC
R _{Gint}	T _j = 25 °C			1.9		Ω
t _{d(on)}	V _{CC} = 600 V	T _j = 125 °C		290		ns
t _r	I _C = 400 A	T _j = 125 °C		60		ns
E _{on}	V _{GE} = +15/-15 V	T _j = 125 °C		39		mJ
t _{d(off)}	R _{G on} = 2 Ω	T _j = 125 °C		670		ns
t _f	R _{G off} = 2 Ω	T _j = 125 °C		80		ns
E _{off}		T _j = 125 °C		64		mJ
R _{th(j-c)}	per IGBT				0.055	K/W



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Typical Applications*

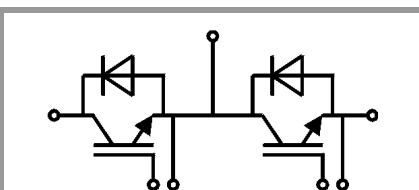
- AC inverter drives
- UPS
- Electronic welders

Remarks

- $I_{DC} \leq 500A$ for $T_{Terminal} = 100\text{ }^{\circ}C$

Characteristics

Symbol	Conditions		min.	typ.	max.	Unit
Inverse diode						
$V_F = V_{EC}$	$I_F = 400\text{ A}$	$T_j = 25\text{ }^{\circ}\text{C}$		1.60	1.80	V
	$V_{GE} = 0\text{ V}$ chiplevel	$T_j = 125\text{ }^{\circ}\text{C}$		1.60	1.80	V
V_{F0}	chiplevel	$T_j = 25\text{ }^{\circ}\text{C}$		1	1.1	V
		$T_j = 125\text{ }^{\circ}\text{C}$		0.8	0.9	V
r_F	chiplevel	$T_j = 25\text{ }^{\circ}\text{C}$		1.50	1.75	mΩ
		$T_j = 125\text{ }^{\circ}\text{C}$		2.00	2.3	mΩ
I_{RRM}	$I_F = 400\text{ A}$	$T_j = 125\text{ }^{\circ}\text{C}$		475		A
Q_{rr}	$di/dt_{off} = 7600\text{ A}/\mu\text{s}$	$T_j = 125\text{ }^{\circ}\text{C}$		96		μC
E_{rr}	$V_{GE} = -15\text{ V}$ $V_{CC} = 600\text{ V}$	$T_j = 125\text{ }^{\circ}\text{C}$		41		mJ
$R_{th(j-c)}$	per diode				0.125	K/W
Module						
L_{CE}				15		nH
$R_{CC'+EE'}$	terminal-chip	$T_C = 25\text{ }^{\circ}\text{C}$		0.35		mΩ
		$T_C = 125\text{ }^{\circ}\text{C}$		0.5		mΩ
$R_{th(c-s)}$	per module			0.02	0.038	K/W
M_s	to heat sink M6		3		5	Nm
M_t		to terminals M6	2.5		5	Nm
						Nm
w					325	g



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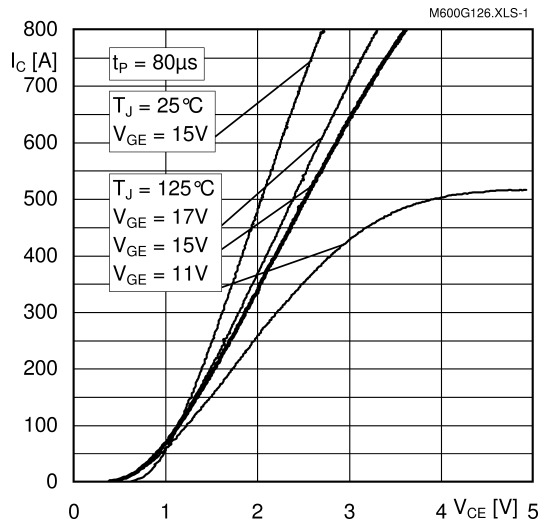


Fig. 1: Typ. output characteristic, inclusive $R_{CC} + EE'$

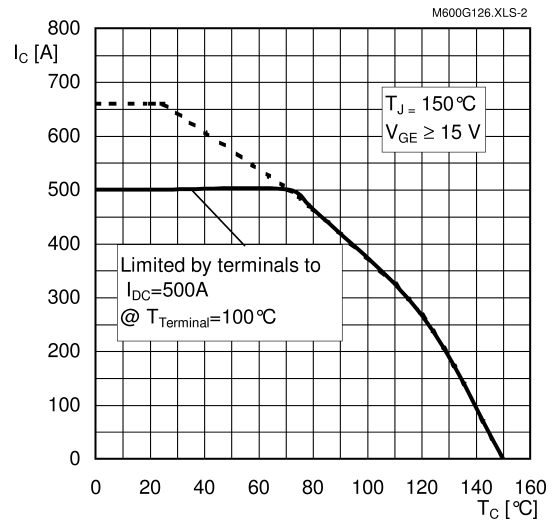


Fig. 2: Rated current vs. temperature $I_C = f(T_C)$

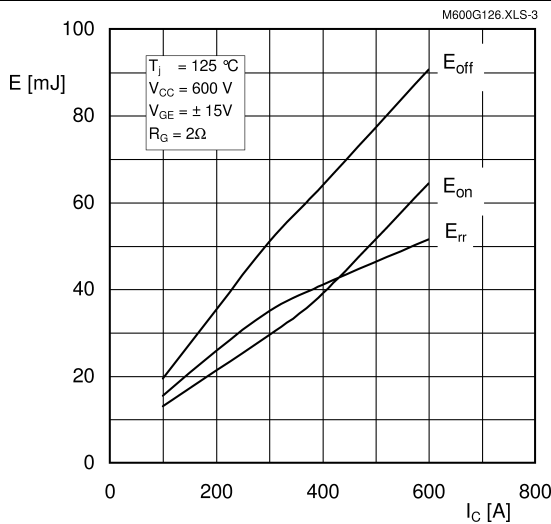


Fig. 3: Typ. turn-on /-off energy = $f(I_C)$

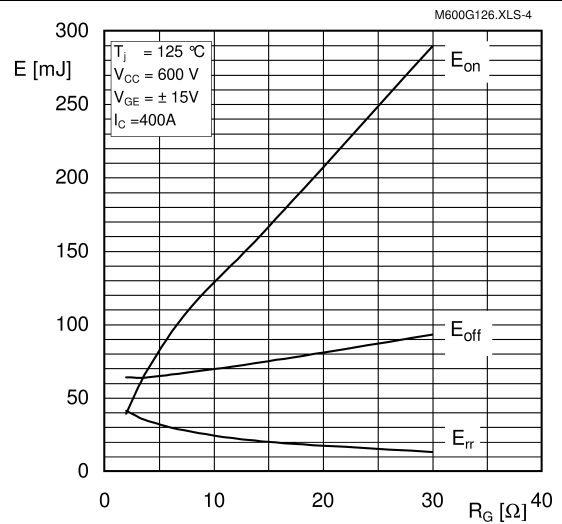


Fig. 4: Typ. turn-on /-off energy = $f(R_G)$

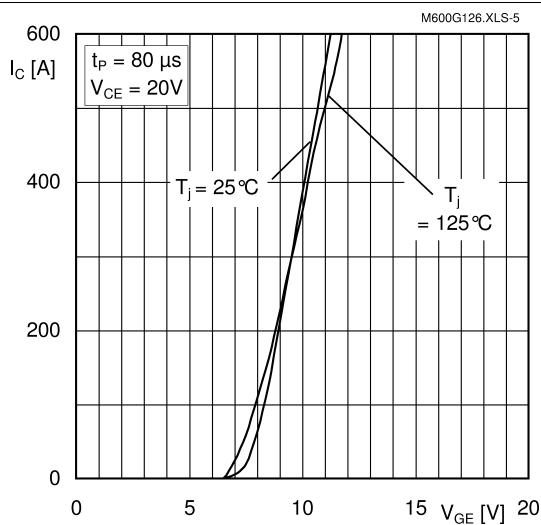


Fig. 5: Typ. transfer characteristic

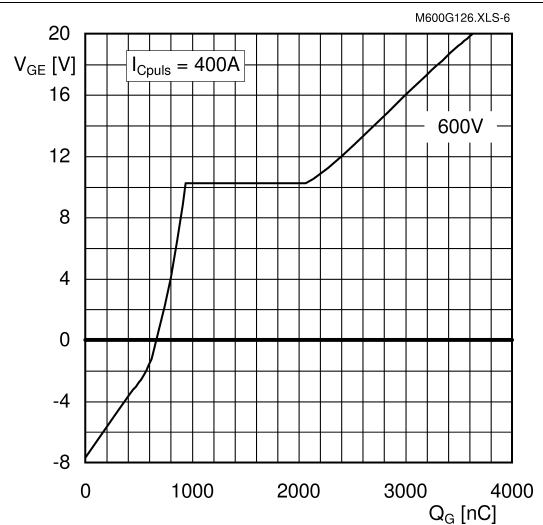


Fig. 6: Typ. gate charge characteristic

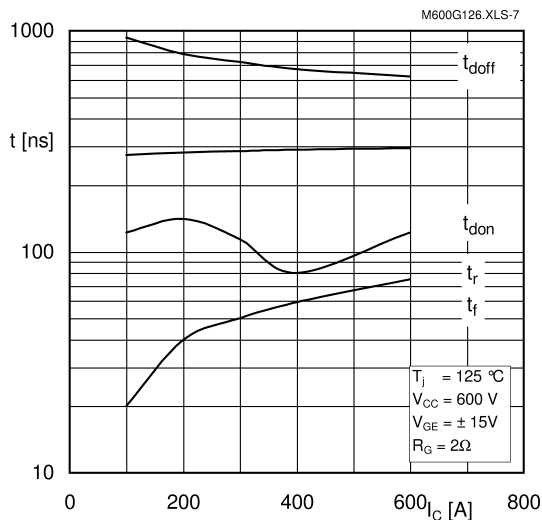


Fig. 7: Typ. switching times vs. I_C

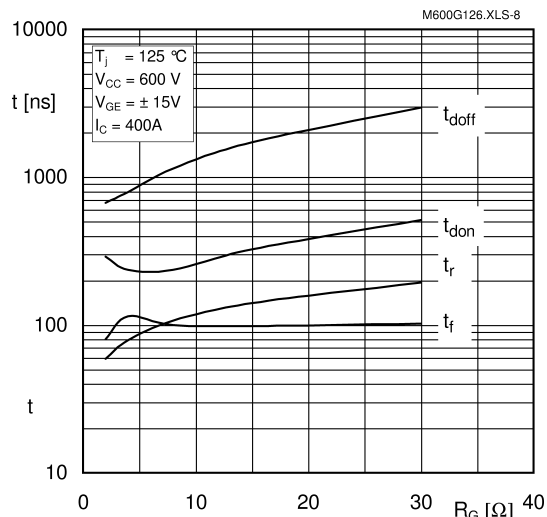


Fig. 8: Typ. switching times vs. gate resistor R_G

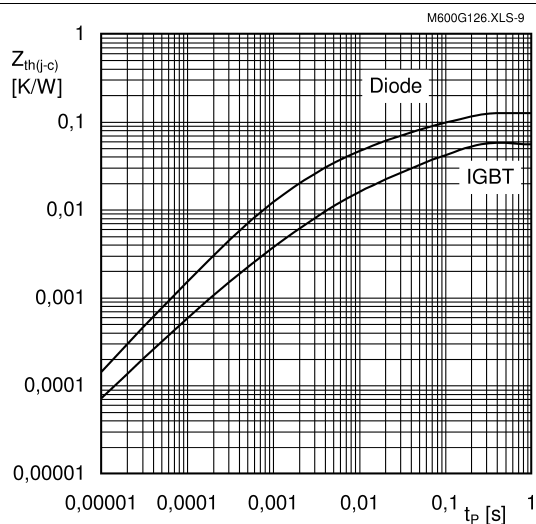


Fig. 9: Transient thermal impedance

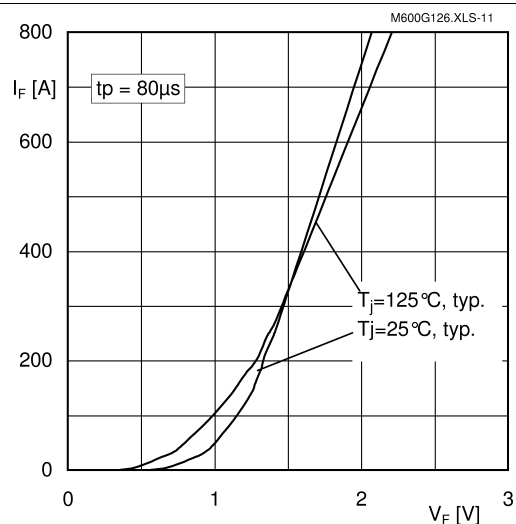


Fig. 10: Typ. CAL diode forward charact., incl. $R_{CC}'+EE'$

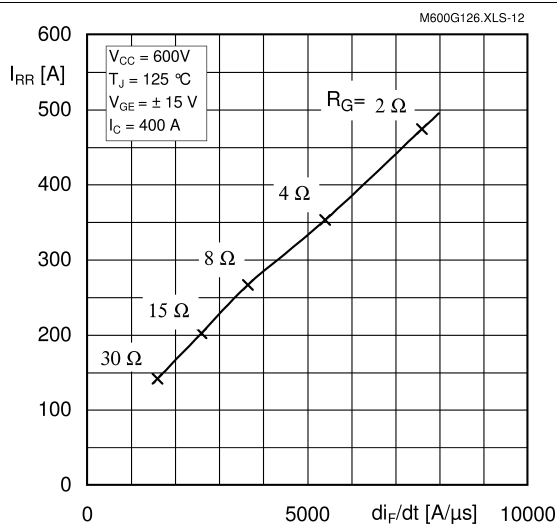


Fig. 11: Typ. CAL diode peak reverse recovery current

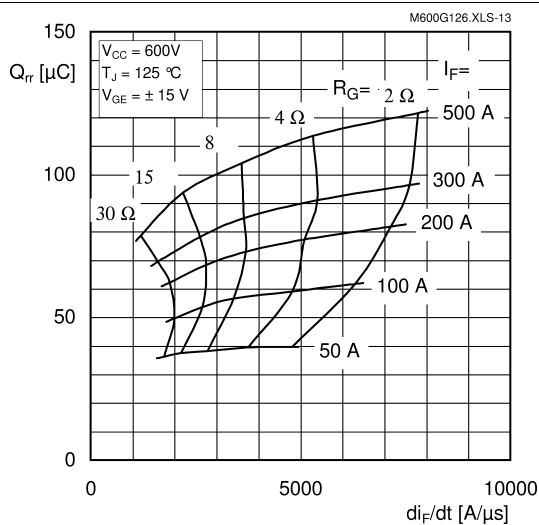
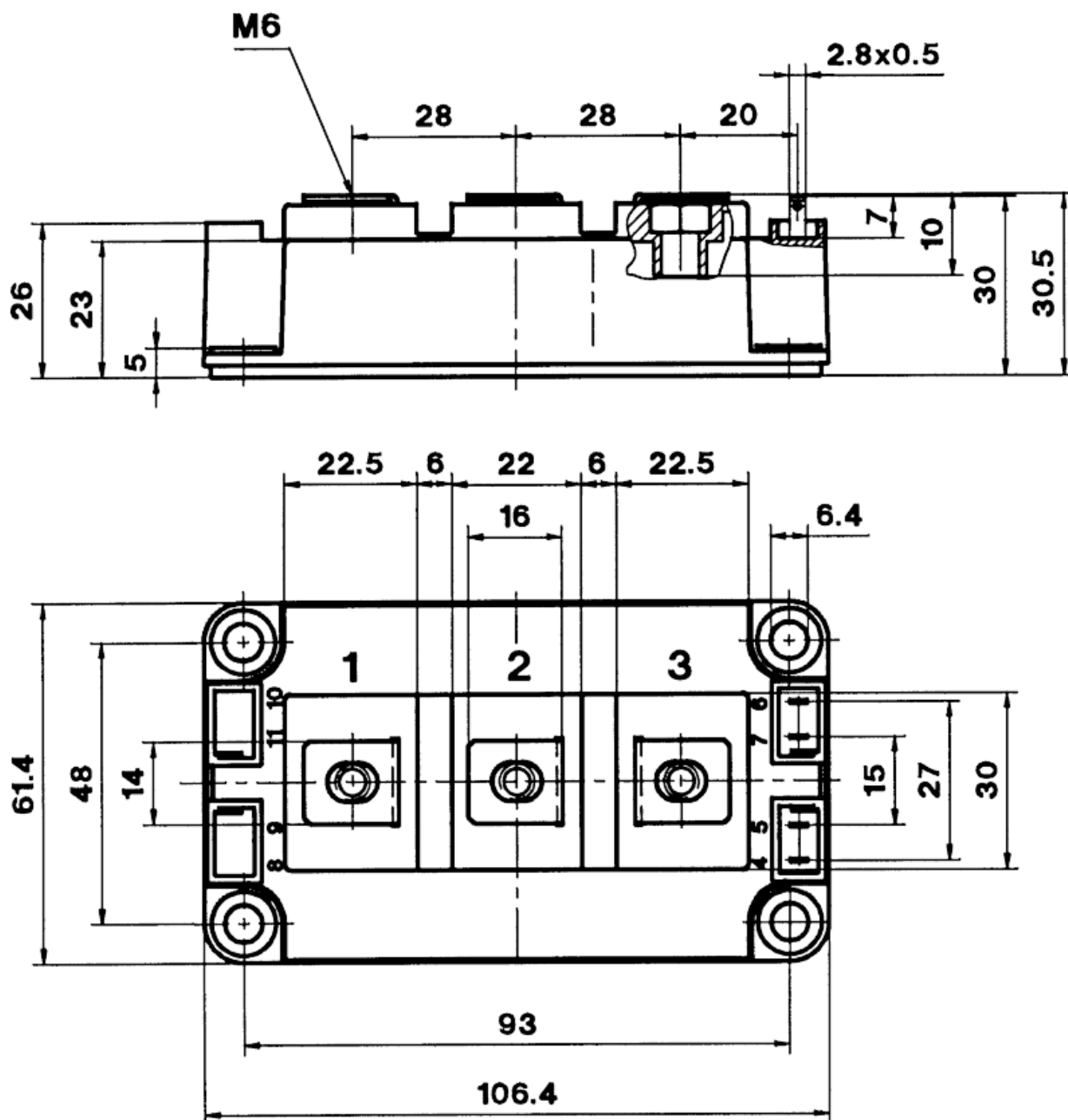
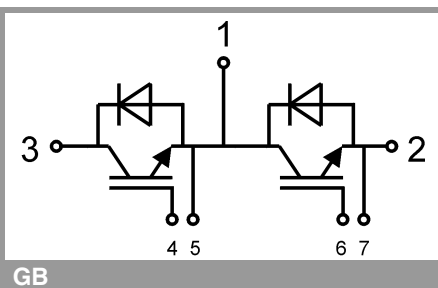


Fig. 12: Typ. CAL diode peak reverse recovery charge



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This is an electrostatic discharge sensitive device (ESDS), international standard IEC 60747-1, Chapter IX

* The specifications of our components may not be considered as an assurance of component characteristics. Components have to be tested for the respective application. Adjustments may be necessary. The use of SEMIKRON products in life support appliances and systems is subject to prior specification and written approval by SEMIKRON. We therefore strongly recommend prior consultation of our staff.